

Adjoint Functors - A Brief Introduction

Ryan Thompson

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Te Herenga Waka - Victoria University of Wellington

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Conversely, given any set S , it is possible to construct the "free" group on S , which in some sense is the most general group one can associate with S .

A basic example of a free group is the integers, which is the unique (up to isomorphism) free group with one generator $(1)!$.

Motivation

In algebra there exists the notion of a **Galois connection**, which is a pair of functions between preordered sets $f : A \rightrightarrows B : g$ such that

$$f(a) \leq b \iff a \leq g(b)$$

for all $a \in A$ and $b \in B$.

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for all $a \in A$ and $b \in B$.

At first glance, this seems like an entirely unrelated concept to the free / faithful relationship - but we will see that these are actually the same category theoretic idea at work.

Prelims

First, we will recall some basic facts about category theory that we will need!

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Definition

A **functor** between two categories C and D is an assignment of an object $F(c) \in C$ for each object $c \in C$, and an arrow $F(f) : Fc \rightarrow Fd$ in D for every arrow $f : c \rightarrow d$ in C . We require that $F(id_c) = id_{Fc}$, and that the following diagram commutes.

$$\begin{array}{ccc} c & \xrightarrow{f} & d \\ \downarrow F & & \downarrow F \\ Fc & \xrightarrow{Ff} & Fd \end{array}$$

A functor of this form is called **covariant**, to emphasize the fact that it preserves the direction of arrows.

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If c, d are objects of a category C , we denote by $\text{hom}(c, d)$ the collection of arrows from c to d . We will assume all of our categories are **small**, such that $\text{hom}(c, d)$ is a **set** (as opposed to a **class**).

Definition

Let C, D be categories, and $F, G : C \rightarrow D$ two functors. A **natural transformation** between F and G , denoted by $F \Rightarrow G$, consists of a collection of arrows $\alpha_c : Fc \rightarrow Gc$ for every object $c \in C$, such that the following **naturality** diagram commutes for any arrow $f : c \rightarrow d$ in C :

$$\begin{array}{ccc} Fc & \xrightarrow{\alpha_c} & Gc \\ \downarrow Ff & & \downarrow Gf \\ Fd & \xrightarrow{\alpha_d} & Gd \end{array}$$

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$$\begin{array}{ccc} Fc & \xrightarrow{\alpha_c} & Gc \\ \downarrow Ff & & \downarrow Gf \\ Fd & \xrightarrow{\alpha_d} & Gd \end{array}$$

We say that a natural transformation is a **natural isomorphism** if each of the arrows α_c above is an isomorphism (i.e. has an inverse).

Example

Consider the category $\mathbf{Set}$ of sets, and the functor $P : \mathbf{Set} \rightarrow \mathbf{Set}$ that maps a set A to its powerset $P(A)$.

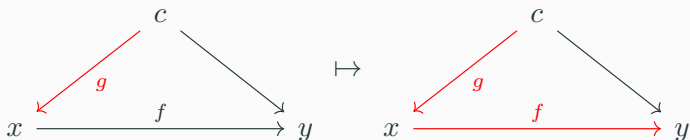
Example

Consider the category $\mathbf{Set}$ of sets, and the functor $P : \mathbf{Set} \rightarrow \mathbf{Set}$ that maps a set A to its powerset $P(A)$.

There is a natural transformation $\mathrm{id}_{\mathbf{Set}} \Rightarrow P$, such that each component $\alpha_A : A \rightarrow P(A)$ maps an element $a \in A$ to the singleton $\{a\} \in P(A)$.

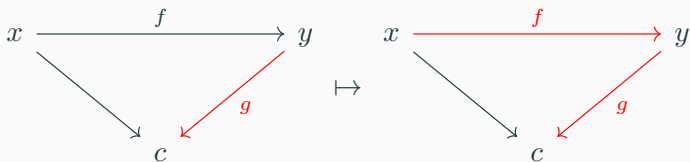
Definition

Let C be a category, and $c \in C$ an object. The **covariant hom functor**, denoted $\text{hom}(c, -)$, maps an object $d \in C$ to the set $\text{hom}(c, d)$, and an arrow $f : x \rightarrow y$ to the post-composition function $f_* : \text{hom}(c, x) \rightarrow \text{hom}(c, y)$ defined by $f_*(g) = f \circ g$.



Definition

The **contravariant hom functor**, denoted $\text{hom}(-, c)$, maps an object $d \in C$ to the set $\text{hom}(d, c)$, and an arrow $f : x \rightarrow y$ to the pre-composition function $f^* : \text{hom}(y, c) \rightarrow \text{hom}(x, c)$, defined by $f^*(g) = g \circ f$.



Adjoints

First Definition

The first way we might define adjoint functors establishes a **natural isomorphism** between particular hom-sets.

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Definition

Let C, D be categories, and $F : C \rightleftarrows D : G$ functors, such that there is a natural isomorphism

$$\mathrm{hom}(Fc, d) \cong \mathrm{hom}(c, Gd)$$

for each $c \in C$ and $d \in D$. Then we say that F, G comprise an **adjunction** between C and D . In particular, we say F is **left-adjoint** to G , and G is **right-adjoint** to F (this is because F appears on the left-side of the hom-functor in the equation above, and G appears on the right!)

We write $F \dashv G$ to indicate that F is left-adjoint to G .

First Definition

The equation above might remind you of the following from functional analysis...

Definition

Let H be a Hilbert space, and A a bounded linear operator. The **adjoint of A** , denoted A^* , is the bounded linear operator satisfying

$$\langle Ax, y \rangle = \langle x, A^*y \rangle,$$

for all $x, y \in H$.

This is where the name "adjoint" in "adjoint functor" came from. However, the A and A^* here aren't "adjoint functors" in any natural sense.

First Definition

This is stating that there is a **one-to-one** correspondence between arrows $f : Fc \rightarrow d$ and $g : c \rightarrow Gd$ that is natural.

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Sometimes, the pair of corresponding morphisms given by this isomorphism are denoted $f^\sharp$ and $f^\flat$.

$$\begin{array}{ccc} Fc & & c \\ \downarrow f^\sharp & \leftrightarrow & \downarrow f^\flat \\ d & & Gd \end{array}$$

Interlude - Whiskering

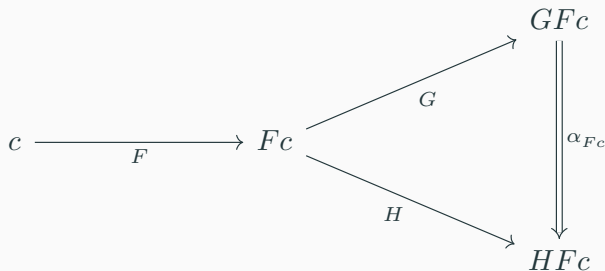
For the second definition of an adjunction, we need to know about **whiskering**!

This is a way to "horizontally" compose a functor with a natural transformation to acquire another natural transformation. More explicitly, let $F : C \rightarrow D$ be a functor, and $G, H : D \rightarrow E$ be functors with a natural transformation $\alpha : G \rightarrow H$ as below.

$$\begin{array}{ccccc} C & \xrightarrow{F} & D & \begin{array}{c} \xrightarrow{G} \\ \Downarrow \alpha \\ \xrightarrow{H} \end{array} & E \end{array}$$

Interlude - Whiskering

We can then define a natural transformation $\alpha F : GF \Rightarrow HF$ with component α_{Fc} at $c \in C$.



Second Definition

Definition

Let C, D be categories, and $F : C \rightleftarrows D : G$ functors, such that there exist natural transformations

$$\eta : 1_C \Rightarrow GF, \quad \epsilon : FG \Rightarrow 1_D$$

such that the following diagrams commute, where $F\eta$ and ϵF are whiskered natural transformations (similarly for ηG and $G\epsilon$).

$$\begin{array}{ccc} F & \xRightarrow{F\eta} & FGF \\ & \searrow \text{id}_F & \downarrow \epsilon F \\ & & F \end{array} \qquad \begin{array}{ccc} G & \xRightarrow{\eta G} & GFG \\ & \searrow \text{id}_G & \downarrow G\epsilon \\ & & G \end{array}$$

For any object $c \in C$, we call the arrow $\eta_c : c \rightarrow GFc$ the **unit** of the adjunction, and for any $d \in D$ we call the arrow $\epsilon_d : FGd \rightarrow d$ the **counit** of the adjunction.

Examples

Example - Free $\dashv$ Forgetful

There are a large class of functors that are "forgetful" in that they "forget" some of the structure or information associated with an object.

e.g. $U : \mathbf{Ab} \rightarrow \mathbf{Set}$ that maps an abelian group to its underlying set. To any set A , we can associate a **free abelian group** that is generated by A modulo the group axioms (i.e. formal $\mathbb{Z}$ -linear combinations of elements of A). Let $F : \mathbf{Set} \rightarrow \mathbf{Ab}$ be the free abelian group functor. Then the pair (U, F) is a free/forgetful adjunction, and U is left-adjoint to F !

Example - Free $\dashv$ Forgetful

We can understand this adjunction by describing its unit and counit. Notice that the functor $UF : \mathbf{Set} \rightarrow \mathbf{Set}$ takes a set A to the set of formal $\mathbb{Z}$ -linear combinations of elements of A (emphasis on **set**).

The component of the unit $\mathrm{id}_{\mathbf{Set}} \Rightarrow GF$ at $A \in \mathbf{Set}$ maps an element $a \in A$ to the trivial combination $a \in UFA$.

Notice that the functor $FU : \mathbf{Ab} \rightarrow \mathbf{Ab}$ takes an abelian group G to the abelian group of formal $\mathbb{Z}$ -linear combinations of elements of the underlying set of G .

The component of the counit $GF \Rightarrow \mathrm{id}_{\mathbf{Ab}}$ takes a formal sum $\sum_{i=0}^n a_i \alpha_i$ of elements of an abelian group G to its evaluation as an element of G .

Interlude - Poset Categories

A poset (short for **partially ordered set**) is a pair $(A, \leq)$ where A is a set, and $\leq$ is a relation on A that is reflexive, antisymmetric and transitive.

Classic example - the powerset of a set X with set inclusion $\subseteq$.

Any poset $(A, \leq)$ can be considered as a category, where the objects are elements of A , and there is an arrow $x \rightarrow y$ if and only if $x \leq y$.

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What does an adjunction look like?

Example - Galois Connection

Let C and D be poset categories. An adjunction between C and D is a pair of functors $F : C \rightleftarrows D : G$ such that

$$Fx \leq y \iff x \leq Gy \quad \text{for all } x \in C, y \in D.$$

In this context, we often refer to F as the **lower (Galois) adjoint** and to G as the **upper (Galois) adjoint** (for obvious reasons)! Furthermore, the pair $F : C \rightleftarrows D : G$ is called a **Galois connection**.

Example - Galois Connection

Let A, B be sets with powersets $P(A)$ and $P(B)$. If $f : A \rightarrow B$ is a function, there are two natural ways to induce a function between $P(A)$ and $P(B)$...

1: the direct image function f^* takes a subset $C \subseteq A$ to the direct image $f(C) \subseteq B$;

2: the preimage function f_* takes a subset $C \subseteq B$ to the preimage $f^{-1}(C) \subseteq A$.

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Thus, the pair $f^* : P(A) \rightleftarrows P(B) : f_*$ form an adjunction! This follows from the fact that if $A' \subseteq A$ and $B' \subseteq B$, then $f(A') \subseteq B'$ if and only if $A' \subseteq f^{-1}(B')$.

This establishes a bijection $\text{hom}(f^*(A'), B') \cong \text{hom}(A', f_*(B'))$ (notice that in a poset category, each hom-set contains either 0 or exactly 1 arrow!)

Monads

A monad on $\mathcal{C}$ is a monoid in the category of endomorphisms on $\mathcal{C}$

We will first see the classic definition, and then an alternative definition via adjunctions.

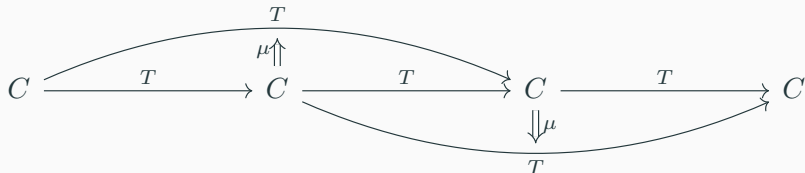
Definition

A **monad** on a category C consists of an endofunctor $T : C \rightarrow C$, along with a natural transformation $\eta : 1_C \Rightarrow T$ called the **unit**, and a natural transformation $\mu : T^2 \Rightarrow T$ such that the following diagrams commute.

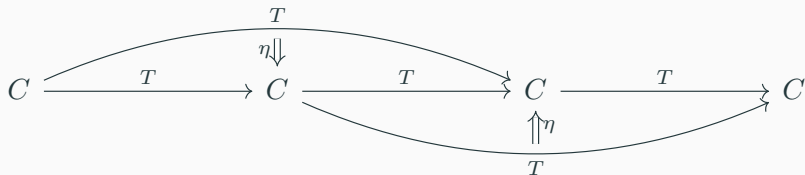
$$\begin{array}{ccc} T^3 & \xrightarrow{T\mu} & T^2 \\ \downarrow \eta T & & \downarrow \mu \\ T^2 & \xrightarrow{\mu} & T \end{array}$$

$$\begin{array}{ccccc} T & \xrightarrow{\eta T} & T^2 & \xleftarrow{T\eta} & T \\ & \searrow \text{id}_T & \downarrow \eta & \swarrow \text{id}_T & \\ & & T & & \end{array}$$

The first diagram amounts to saying there are two ways to whisker μ with T , and they give you the same result.



The second diagram amounts to saying there are two ways to whisker η with T , and they yield the same result.



"All these arrows are nice, but it's just a bunch of abstract nonsense!"

Monads from Adjunctions

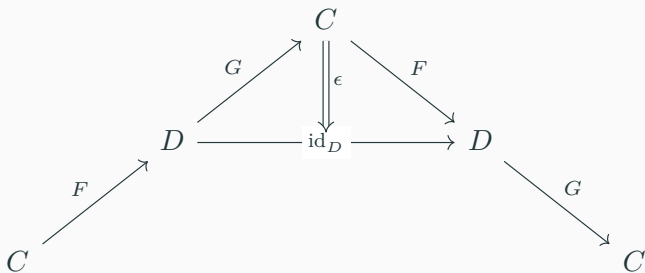
In her book "Category Theory in Context", Emily Riehl says
"Monads are the shadows cast by adjunctions"

Recall that a monad is an endofunctor with additional structure.

Every adjunction $F : C \rightleftarrows D : G$ gives rise to a monad on C given by the endofunctor GF .

The unit $\text{id}_C \Rightarrow GF$ is given by the unit of the adjunction.

The whiskered counit $G\epsilon F : GF GF \Rightarrow GF$ is the multiplication for the monad.



Monads - Example

Consider the free $\dashv$ forgetful adjunction between the categories $\mathbf{Set}$ of sets, and $\mathbf{Set}_*$ of pointed sets (a set with a distinguished element).

The forgetful functor $U : \mathbf{Set}_* \rightarrow \mathbf{Set}$ forgets that the distinguished point is special, while the free functor $F : \mathbf{Set} \rightarrow \mathbf{Set}_*$ maps a set A to the pointed set $A_+ := A \cup \{A\}$.

Monads - Example

The unit of the monad $\eta : \text{id}_C \Rightarrow UF$ is the obvious inclusion $A \hookrightarrow A \cup \{A\}$.

The multiplication component $\mu : UFUF \Rightarrow UF$ at an object

$$(A_+)_+ := (A \cup \{A\}) \cup \{A \cup \{A\}\}$$

behaves as follows:

$$\mu_{(A_+)_+}(a) = \begin{cases} a & \text{if } a \in A, \\ \{A\} & \text{otherwise.} \end{cases}$$

This monad on Set is called the **maybe** monad - in certain contexts we want to consider total maps $f : A \rightarrow B_+$, where the preimage of the distinguished point of B_+ are considered to be the points in A on which f is "undefined".

Monads - My Favourite Example!

Let C, D be poset categories, and $F : C \rightleftarrows D : G$ an adjunction, such that GF is a monad on C . What does this mean?

It means that $c \leq GFc$, and $GF GFc \leq GFc$ - thus $GF GFc = GFc$ - for all $c \in C$. This is exactly what a closure operator is!

For instance, let C be the poset of open subsets of a topological space X . If F is a monad on C , then for all $U \subseteq X$, we have $U \subseteq FU$, and $F^2U = FU$.

Under this definition, the operation "closure of an open set" in topology is a monad on its poset of open sets! (very cool)